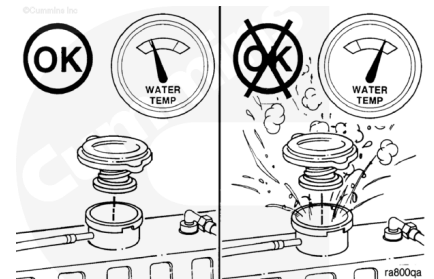


Trainings ▾

Drain

⚠ WARNING ⚠

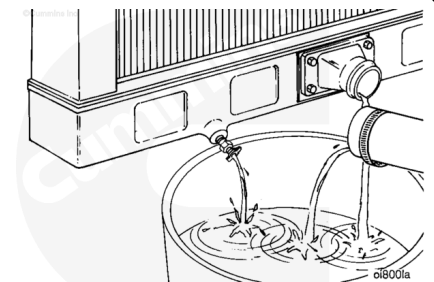
Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Remove the radiator cap after the engine is cool.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



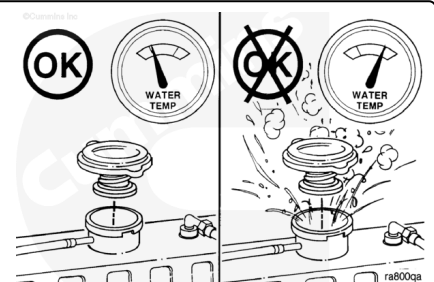
Position the vehicle on level ground. Open the draincock at the bottom of the radiator. Remove the lower radiator hose. Drain the cooling system.

Do **not** remove the coolant filter.

Flush

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



The performance of RESTORE™ is dependent on time, temperature, and concentration levels. An extremely scaled or flow restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning time. RESTORE™ can be safely used up to twice the recommended concentration levels. Extremely scaled or fouled systems can require more than one cleaning.

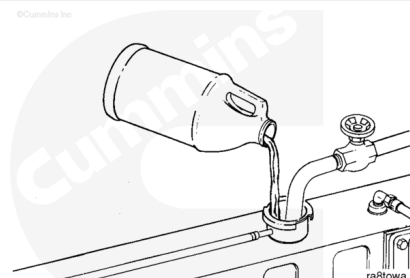


⚠ CAUTION ⚠

Fleetguard® RESTORE™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

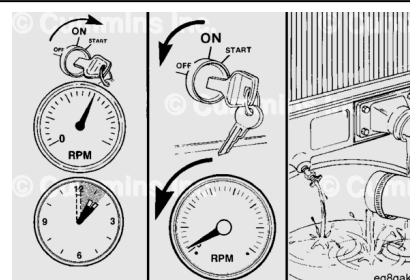
Immediately add 3.8 liters [1 gal] of Fleetguard® RESTORE™, or equivalent, for each 38 to 57 liters [10 gal] of cooling system capacity, and fill the system with plain water.

Turn the heater temperature switch to HIGH to allow maximum coolant flow through the heater core. The blower does **not** have to be in the ON position.

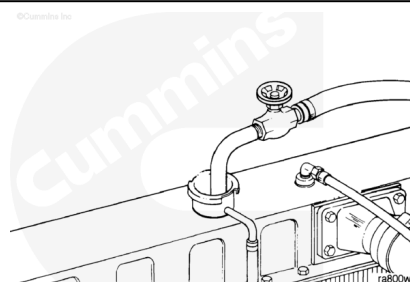


Operate the engine at a coolant temperature of at least 85°C [185°F] for 1 to 1½ hours.

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.



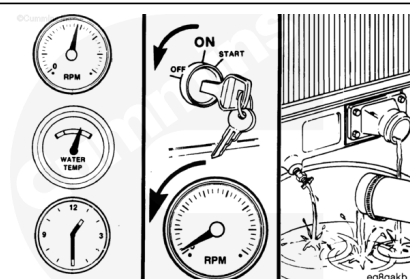
Fill the cooling system with clean water.



Operate the engine at high idle for five minutes with the coolant temperature above 85°C [185°F].

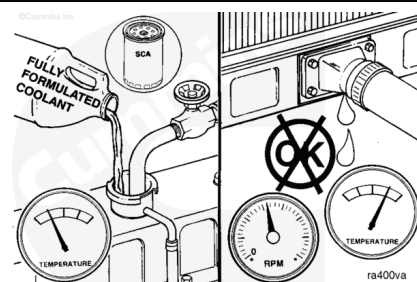
Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.

If the water drained is still dirty, the system **must** be flushed again until the water is clean.

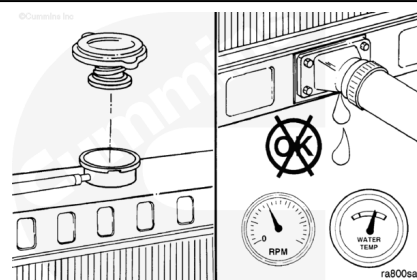


Install a new coolant filter and fill the cooling system with fully formulated coolant.

Use additional SCA to bring the coolant to the correct SCA concentration level. See the specifications in this section.



Install the radiator cap. Operate the engine until the coolant reaches a temperature of 70°C [160°F]. Check for coolant leaks.

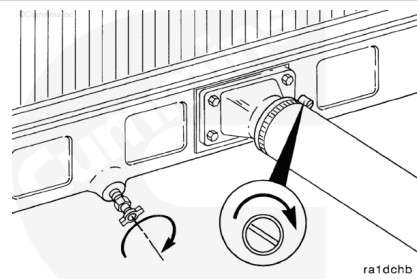


Fill

Close the radiator draincocks.

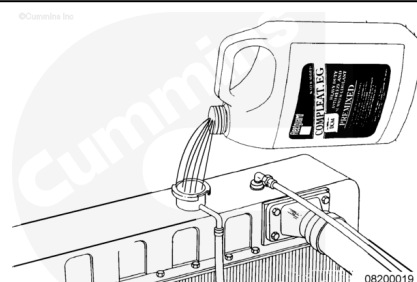
Install the lower radiator hose(s). Tighten the hose clamps.

Torque Value: 5 n•m [40 in-lb]

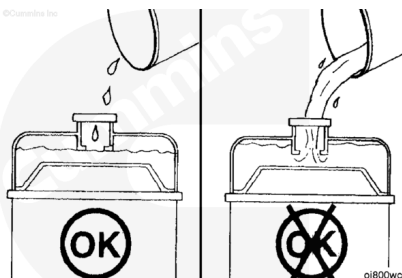


Cummins Inc. recommends using either a 50/50 mixture of good quality water and fully formulated antifreeze, or fully formulated coolant when filling the cooling system.

Use the correct units of SCA to obtain the correct cooling system protection.

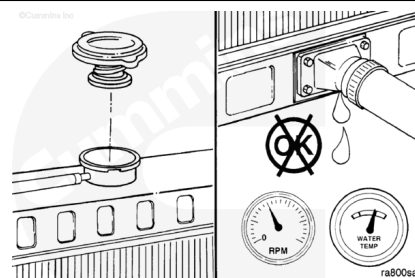


Fill the cooling system with coolant to the bottom of the fill neck in the radiator fill (or expansion) tank.



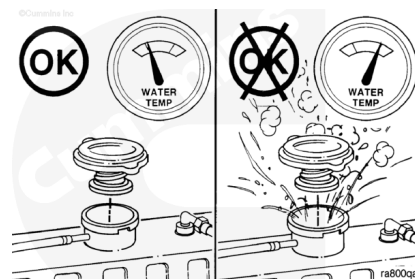
Replace the radiator cap.

Operate the engine until the coolant reaches a temperature of 70°C [160°F]. Check for leaks.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Shut the engine OFF. Allow it to cool.

Check the coolant level. Fill if necessary.

Clean

⚠ CAUTION ⚠

Do not use caustic cleaners in the cooling system. Aluminum components can be damaged.

Change the coolant and antifreeze every 6000 hours or two years. See Cummins Requirements for Cooling System Extended Requirements for Cooling System Extended Service Intervals, Bulletin 3666286.

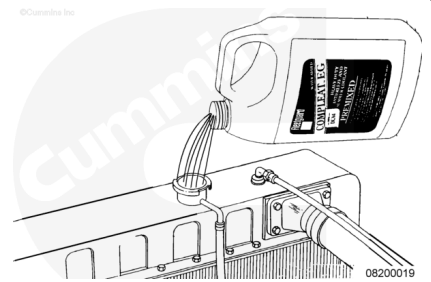
The cooling system **must** be clean to work correctly. Drain the system and flush with clean water. If the system shows mineral buildup, scale, rust, or oil, clean with a heavy-duty



engine coolant cleaner and follow manufacturer's instructions.

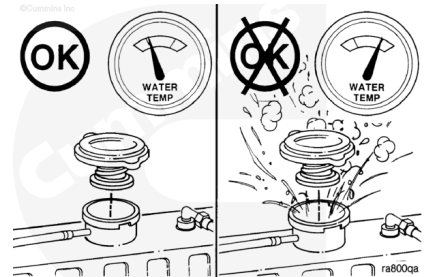
Fill the cooling system with the correct mixture of antifreeze and water, and the correct SCA, or equivalent units. Refer to Procedure 008-046 in Section 4.

(/qs3/pubsys2/xml/en/procedures/99/99-008-046.html)



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Operate the engine and check for coolant leaks.

After the air has been purged from the system, check the coolant level again.